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Comparison on characterization and antioxidant activity of polysaccharides from *Ganoderma lucidum* by ultrasound and conventional extraction

Qiaozhen Kang ^a, Sisi Chen ^a, Shufang Li ^a, Bo Wang ^a, Xin Liu ^a, Limin Hao ^b  , Jike Lu ^a  

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Highlights

- Polysaccharides from *Ganoderma lucidum* (GLP_{UAE} and GLP_{HWE}) were obtained.
- Characterization and antioxidant activity of polysaccharides were investigated.
- GLP_{UAE} and GLP_{HWE} were purified by chromatographic column.
- The antioxidant activity of GLP_{HWE} was better than GLP_{UAE}.

Abstract

The polysaccharides of *Ganoderma lucidum* (GLP) extracted by the methods of ultrasound assisted extraction (UAE) and hot water extraction (HWE) were characterized and the antioxidant activities

of GLP_{UAE} and GLP_{HWE} were compared. High performance gel permeation chromatography (HPGPC) showed that the average molecular weight of GLP_{UAE} and GLP_{HWE} were 465.65 kDa and 703.45 kDa, respectively. GLP_{UAE} was composed of mannose, rhamnose, glucose, galactose and arabinose in the molar ratio of 2.58:1.25:11.17:2.5:1, while GLP_{HWE} was composed of the same monosaccharide in the ratio of 3.11:1.11:19.44:2.33:1. GLP_{HWE} showed a relatively higher antioxidant activity than GLP_{UAE} by testing the reducing power, the scavenging ability on 1,1-diphenyl-2-picryl-hydrazyl (DPPH), hydroxyl radical and cellular protective effect on yeast cells from ultraviolet radiation (UV) damage. GLP_{UAE} and GLP_{HWE} were purified by chromatographic column of DEAE-52 cellulose. GLP_{UAE}, GLP_{HWE} and the purified components could be novel antioxidants for functional food.

Introduction

Ganoderma lucidum (Leyss. ex Fr.) Karst, is a species of basidiomycetes which belongs to Polyporaceae of Aphyllophorales, and it is one of the most popular medicinal mushrooms in Asian countries. *Ganoderma lucidum* has been under modern biochemical and pharmacological research during the last 30 years. Previous literatures showed that *Ganoderma lucidum* could promote health and longevity due to its numerous bioactive components such as polysaccharides, triterpenoids, ergosterols, various proteins, unsaturated fatty acids, vitamins and minerals [1].

Ganoderma lucidum polysaccharides (GLP) are the primary bioactive components, and they have been demonstrated to possess diverse and significant potential pharmacological activities such as antitumor, immunomodulatory and antioxidant activities have been introduced in the market. GLP have exhibited strong antioxidant properties and might be a resource for natural antioxidants exploration [2,3]. Nowadays, more and more polysaccharide products as functional foods have been studied.

GLP are mainly extracted from the fruiting bodies of *Ganoderma lucidum*. The conventional extraction method is hot water extraction (HWE). However, it should be noted that HWE of polysaccharides has disadvantages of long extraction time and high extraction temperature [4]. Ultrasound assisted extraction (UAE) is a new developing method with many advantages, such as economizing power consumption and shortening extraction duration [5,6]. The potential bioactivity of polysaccharides from *Ganoderma lucidum* may be affected by different extraction methods. To evaluate this hypothesis, the aim of this work was to investigate the characterization and antioxidant activity of GLP_{UAE} and GLP_{HWE}. Then GLP_{UAE} and GLP_{HWE} were purified using DEAE-52 cellulose chromatography to obtain purified fractions.

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Materials and chemicals

Fruiting bodies of *Ganoderma lucidum* were collected from Qiaokang Pharmaceutical Co., Ltd. (Jinzhai, Anhui province, China). 1-Phenyl-3-methyl-5-pyrazolone (PMP) and trifluoroacetic acid (TFA) were obtained from Aladdin (Shanghai, China). Galactose (Gal), arabinose (Ara), rhamnose (Rha) and mannose (Man) were obtained from Sigma-Aldrich Co. (St. Louis, USA). All other chemicals were of analytical grade. ...

Extraction, isolation and purification of polysaccharides

The UAE of GLP was carried out using an Ultrasonic Cleaner (KQ-250DE, Kun Shan Ultrasonic ...

Molecular weight determination

The main components of GLP_{UAE} and GLP_{HWE} were detected through a gel permeation chromatographic column. The elution peaks of both GLP_{UAE} (Fig. 1A) and GLP_{HWE} (Fig. 1B) were multiple. There were four peaks in the elution profiles of GLP_{UAE} at 10.89, 14.75, 15.48 and 18.80 min, while the corresponding molecular weight were calculated to be 1349 kDa, 110 kDa, 68 kDa and 7 kDa, respectively. The relative contents of the corresponding four polysaccharides were 31.81%, 16.7%, 23.85% and 27.65%, ...

Conclusions

Comparison of characterization of GLP_{UAE} and GLP_{HWE} was conducted in this study. Results from HPGPC showed that the average molecular weight of GLP_{UAE} and GLP_{HWE} were 465.65 kDa and 703.45 kDa, respectively. The monosaccharide component determination results manifested that the composition of both GLP_{UAE} and GLP_{HWE} were almost similar, with just some differential in the molar ratio. Comparison of antioxidant activity of the two polysaccharides showed that GLP_{HWE} exhibited stronger abilities of ...

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Conflict of interest statement

The authors declare no conflict of interest. ...

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...A comparison of the *Ganoderma lucidum* polysaccharides that were extracted using a variety of different methods was performed, and the authors stated that lower molecular weight was obtained using UAE. Although the polysaccharides displayed the same monosaccharide compositions across the differing methods, the monosaccharide ratio was changed by US, and UAE decreased the reducing power of the polysaccharides [35]. As far as fatty acids, oils and lipids are concerned, an Iranian research group has recently investigated the effectiveness of the UAE treatment of *Aesculus hippocastanum* fruit oil, as optimised by response surface methodology using BBD, and compared it to the traditional Soxhlet method....

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